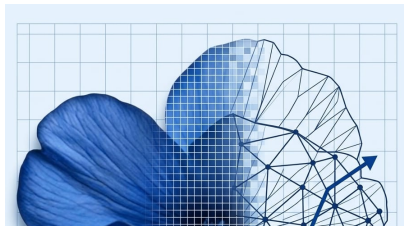


NORMALIZED CUTS & IMAGE SEGMENTATION

Jianbo Shi and Jitendra Malik (2000)



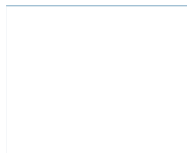
A New Perspective on Segmentation

- The paper proposes a novel global approach to solve the perceptual grouping problem in vision by capturing an image's overall structure rather than focusing on local features.
- Instead of focusing on local pixel features, Shi and Malik emphasize on extracting the global impression of the image.
- Image segmentation is treated as a graph partitioning problem using a new global criterion called the **normalized cut**.
- This measures **both, the** total dissimilarity between the different groups as well as the total similarity within the groups.
- Along with this, they have also shown an efficient computational technique based on generalized eigenvalue problem that can be used to optimize this criterion.

Grouping as graph partitioning

Graph Representation

- An image is treated as a weighted undirected graph $G = (V, E)$.
 - **Nodes** (V): Pixels of the image.
 - **Edges** (E): Connections between every pair of nodes.
 - **Weight** (w_{ij}): Similarity between nodes i and j based on spatial proximity and feature (intensity, color, brightness etc).



$$\text{if } \|X(i) - X(j)\| \leq r$$

otherwise

$$\begin{cases} e^{\frac{-\|F(i)-F(j)\|^2}{\sigma_F^2}} \cdot e^{\frac{-\|X(i)-X(j)\|^2}{\sigma_X^2}} \\ 0 \end{cases}$$

THE PARTITIONING GOAL

The core objective is to partition the graph into disjoint sets such that the similarity within sets (intra-cluster) is high, and the similarity across different sets (inter-cluster) is low.

$$w(i, j) = \begin{cases} e^{-\frac{\|F(i)-F(j)\|^2}{\sigma_F^2}} \cdot e^{-\frac{\|X(i)-X(j)\|^2}{\sigma_X^2}} & \text{if } \|X(i) - X(j)\| < r, \\ 0 & \text{otherwise.} \end{cases}$$

where $X(i)$: spatial location of i , $F(i)$: feature of i .

The Min-Cut

- Traditional minimum cut algorithms aim to minimize **cut cost**.
- However, this heavily biases the algorithm toward cutting small sets of isolated nodes, failing to capture the global structure of the image.

$$\text{Cut}(A, B) = \sum_{u \in A, v \in B} w(u, v)$$

Optimal cut Min cut result

$$\text{Cut}(A, B) = \sum_{u \in A, v \in B} w(u, v)$$

- Min-Cut favors cutting small sets of isolated nodes, as the absolute sum of edge weights is naturally smaller for fewer nodes.
- This fails to capture the global structure of the image which makes the final segmentation absolutely meaningless.

The Mathematical Solution

NCut: Normalized Cut Value — Balancing the Cut

- To avoid unnatural bias for partitioning out small regions, Shi and Malik proposed the Normalized Cut (NCut). It computes the cut cost as a fraction of the total edge connections to all nodes in the graph.
- $\text{assoc}(A, V)$: the total connection from nodes in A to all nodes in the entire graph.
- $\text{assoc}(B, V)$: the total connection from nodes in B to all nodes in the entire graph.
- Minimizing **NCut** is equivalent to increasing the **association**, thus making sure that we don't isolate a single pixel.

$$\text{Ncut} (A , B) = \frac{\text{cut} (A , B)}{\text{assoc} (A , V)} + \frac{\text{cut} (A , B)}{\text{assoc} (B , V)}$$

The Mathematical Solution

Minimizing the dissimilarity between partitions is mathematically equivalent to maximizing the within-cluster similarity.

Normalized Association

$$N_{\text{assoc}}(A, B) = \frac{\text{assoc}(A, A)}{\text{assoc}(A, V)} + \frac{\text{assoc}(B, B)}{\text{assoc}(B, V)}$$

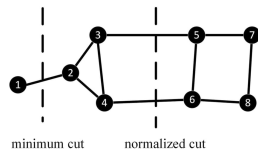
Relation between normalized cut and normalized association

$$N_{\text{cut}}(A, B) + N_{\text{assoc}}(A, B) = 1$$

- **$N_{\text{cut}}(A, B)$ decreases as $N_{\text{assoc}}(A, B)$ increases.**
- $\text{assoc}(A, A)$ and $\text{assoc}(B, B)$ are total weights of edges connecting nodes within A and B respectively.

$$N_{\text{assoc}}(A, B) = \frac{\text{assoc}(A, A)}{\text{assoc}(A, V)} + \frac{\text{assoc}(B, B)}{\text{assoc}(B, V)}$$

$$N_{\text{Cut}}(A, B) = 2 - N_{\text{assoc}}(A, B)$$



Mathematical Formulation

Minimizing **Ncut** is an **NP hard problem**. However, it can be transformed into a generalized eigenvalue system.

W: Affinity matrix (similarities).

$$w_{ij} = \begin{cases} e^{-\frac{\|F(i)-F(j)\|^2}{\sigma_f^2}} \cdot e^{-\frac{\|X(i)-X(j)\|^2}{\sigma_x^2}} & \text{if } \|X(i) - X(j)\| < r, \\ 0 & \text{otherwise.} \end{cases}$$

D: Diagonal matrix.

$$D_{ii} = d_i = \sum_j w_{ij}$$

The second smallest eigenvector of this system provides us the optimal partition.

$$(D - W) y = \lambda D y$$

$$\begin{cases} e^{-\frac{\|F(i)-F(j)\|^2}{\sigma_f^2}} * e^{-\frac{\|X(i)-X(j)\|^2}{\sigma_x^2}} \\ 0 \end{cases}$$

$$\text{if } \|X(i) - X(j)\| \leq r \quad \begin{matrix} D: \text{Diagonal matrix} \\ D_{ii} = d_i = \sum_{j=1}^N w_{ij} \end{matrix}$$

otherwise

The Grouping Algorithm — Step 1: Construct the graph

- Construct a weighted graph $G(V, E)$ by treating each pixel as a node and connecting them by weighted edges that represent the likelihood that they belong to the same object.
- The edge weight is determined on the basis of the similarity (intensity, color, texture etc.) between the pixels and their spatial proximity.

The Grouping Algorithm — Step 2: Construct the matrices

- Construct the affinity matrix W and the diagonal matrix D .
- This is done by translating the information of edge weights.

The Grouping Algorithm — Step 3: Solve eigen system

- Solve for the eigenvalues of the system below and compute the eigenvectors.

$$(D - W)y = \lambda D y$$

- Extract the eigenvector corresponding to the second smallest eigenvalue.

Solve for the eigenvalues of
the system $(D-W)y=\lambda D y$ and
compute the eigen vectors

The Grouping Algorithm — Step 4: Split graph

- Once the eigenvectors are computed, we can partition the graph into two pieces using the second smallest eigenvector.
- This is done by finding a splitting point (usually 0) and assigning pixels to one group or the other with respect to it.

The Grouping Algorithm — Step 5: Recursive cut

- After the graph is broken into two pieces, calculate the **Ncut** value for the current bipartition; we can recursively run our algorithm on the two partitioned parts.
- This is continued till **$\text{Ncut}(A,b) \geq \text{threshold value}$** .
- (i.e., the connection between A and B is nearly as strong as their internal volume, thus no clear partitions can be found).

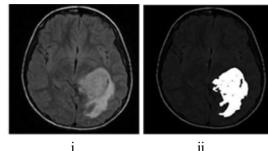
Ncut Results



Source: Berkeley segmentation dataset

Practical Applications

- **Medical Imaging:** Crucial for accurate organ and tumor boundary extraction in complex MRI and CT scans where edges are ambiguous.
- **Autonomous Driving:** Used in scene parsing to differentiate between drivable roads, pedestrians, and obstacles.
- **Interactive Photo Editing:** Enabling intelligent foreground-background separation tools in design software.



- This paper has developed a grouping algorithm based on the view that perceptual grouping should be a process that aims to extract global impressions of a scene and provide a hierarchical description of it.
- A computational method based on this idea has been developed and applied to segmentation of brightness, colour and texture.
- Results of experiments on real and synthetic images are very encouraging and illustrate that the normalized cut criterion does indeed satisfy our initial goal of extracting the “big picture” of a scene.



Jianbo Shi and Jitendra Malik, *Normalized Cuts and Image Segmentation*, IEEE Transactions on Pattern Analysis and Machine Intelligence (PAMI), Vol. 22, No. 8, pp. 888–905, August 2000.

Thank You